

Claude MCP Server on Kali Linux

Installation, Configuration & Troubleshooting Guide

Internal Lab Reference Document

All Rights Reserved © N.M.I Raisul Bari

Scope: mcp-kali-server + Claude Desktop integration on Kali Linux,
routed lab network (Kali ↔ OPNsense ↔ Metasploitable2)

Authorized lab/training environment use only

All rights reserved by N.M.I Raisul Bari

Table of Contents

1. Overview & Architecture	3
1.1 Components	3
1.2 High-Level Diagram	3
2. Prerequisites	4
3. Installation	5
3.1 Install mcp-kali-server	5
3.2 Install Claude Desktop	5
3.3 Install the Underlying Security Tools	5
4. Configuration	6
4.1 Register the MCP Server in Claude Desktop	6
4.2 Reference claude_desktop_config.json	6
5. Known Issues & Fixes	7
5.1 CommandExecutor “expects a string, but got list”	7
5.2 Claude Desktop crashes: “Trace/breakpoint trap” (exit 133)	7
5.3 “Connection refused” on localhost:5000	8
5.4 Malformed claude_desktop_config.json	8
6. Lab Network Configuration (Kali ↔ OPNsense ↔ Metasploitable2)	9
6.1 Addressing Plan	9
6.2 Create the Second Host-Only Network	9
6.3 Configure Kali	9
6.4 Configure OPNsense	9
6.5 Configure Metasploitable2	10
6.6 Verification	10
7. Usage Examples	11
8. Security & Operational Notes	12

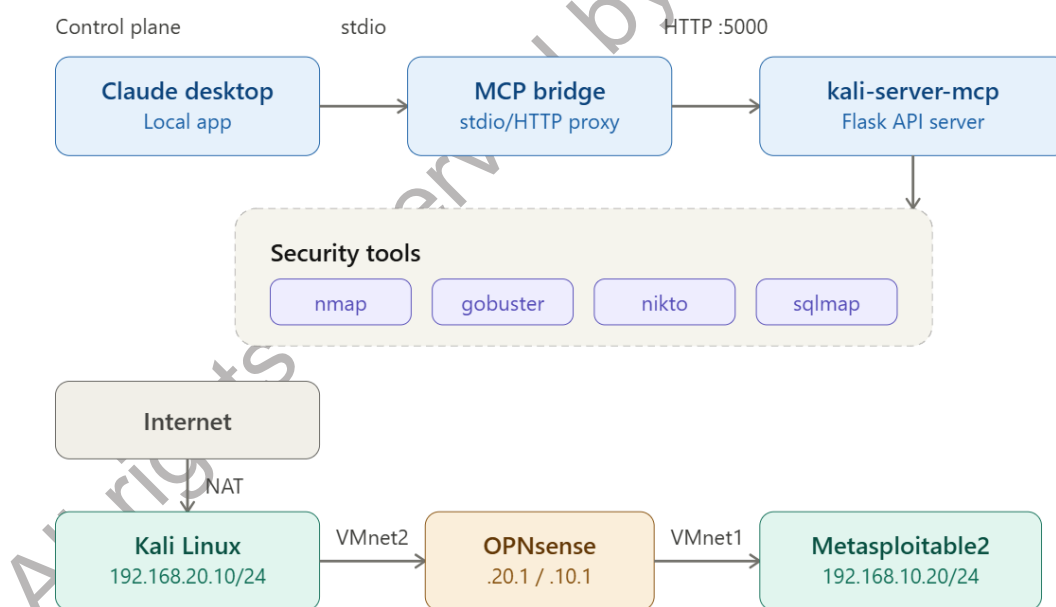
1. Overview & Architecture

This document covers the end-to-end setup connecting Claude Desktop to Kali Linux security tools via an MCP (Model Context Protocol) server, and the supporting lab network used to test against Metasploitable2 through an OPNsense firewall.

1.1 Components

- Kali Linux – host running Claude Desktop, mcp-kali-server, and the security tools themselves
- mcp-kali-server – local Flask API (kali-server-mcp) plus an MCP bridge (mcp-server) that exposes nmap, gobuster, dirb, nikto, sqlmap, metasploit, hydra, john, wpscan, and enum4linux as callable tools
- Claude Desktop – community-maintained Linux build (aaddrick/claude-desktop-debian), configured with a Local MCP Server entry pointing at the bridge
- OPNsense – routes and firewalls traffic between the Kali-facing network and the Metasploitable2-facing network
- Metasploitable2 – intentionally vulnerable target VM, isolated on its own host-only network

1.2 High-Level Diagram



Note: keep this entire stack inside an isolated lab network. Only point these tools at systems you own or are explicitly authorized to test.

2. Prerequisites

- Kali Linux VM with internet access (for package installation)
- Root or sudo access on the Kali VM
- A hypervisor supporting multiple virtual networks (VMware Workstation/Player used in this guide)
- Metasploitable2 VM, deployed on an isolated host-only network
- OPNsense VM with at least two configurable network interfaces

All rights reserved by N.M.I Raisul Bari

3. Installation

3.1 Install mcp-kali-server

Installs the Flask API server and MCP bridge that exposes Kali tools to Claude.

```
sudo apt update
sudo apt install -y mcp-kali-server
```

Start the API server once to confirm it runs (Ctrl+C to stop – this is just a smoke test):

```
kali-server-mcp
```

Confirm the bridge binary and check whether it needs an explicit API address:

```
which mcp-server
mcp-server --help
```

3.2 Install Claude Desktop

There is no official Anthropic Linux build of Claude Desktop. This guide uses the community-maintained aaddrick/claude-desktop-debian repository.

```
curl -fsSL https://pkg.claude-desktop-debian.dev/KEY.gpg | sudo gpg --dearmor -o
/usr/share/keyrings/claude-desktop.gpg
echo "deb [signed-by=/usr/share/keyrings/claude-desktop.gpg arch=amd64,arm64]
https://pkg.claude-desktop-debian.dev stable main" | sudo tee
/etc/apt/sources.list.d/claudedesktop.list
sudo apt update
sudo apt install -y claude-desktop
```

Run the built-in diagnostic before first launch to catch sandbox/permission issues early:

```
claude-desktop --doctor
```

3.3 Install the Underlying Security Tools

mcp-kali-server only provides the API wrapper – the actual tools must be installed separately:

```
sudo apt install -y dirb gobuster nikto nmap enum4linux-ng \
hydra john metasploit-framework sqlmap wpscan wordlists
sudo gunzip /usr/share/wordlists/rockyou.txt.gz
```

4. Configuration

4.1 Register the MCP Server in Claude Desktop

1. Open Claude Desktop → Settings → Developer → Local MCP servers.
2. Click “Edit Config” – this opens `claude_desktop_config.json` in your default editor.
3. Add an `mcpServers` entry for the Kali bridge (see 4.2 for the full file).
4. Save, fully quit Claude Desktop (from the tray/menu, not just the window), and relaunch.
5. Open a new chat and check the hammer/tools icon at the bottom of the message box – it should list `kali-mcp` and its tools.

4.2 Reference `claude_desktop_config.json`

Path: `~/.config/Claude/claude_desktop_config.json`

```
{
  "preferences": {
    "coworkHipaaRestricted": false,
    "coworkWebSearchEnabled": true,
    "coworkModelAutoFallbackByAccount": {
      "<your-account-uuid>": true
    },
    "remoteToolsDeviceName": "kali",
    "coworkScheduledTasksEnabled": false,
    "ccdScheduledTasksEnabled": false,
    "bypassPermissionsGateByAccount": {
      "<your-account-uuid>": false
    },
    "epitaxyPrefs": {
      "starred-local-code-sessions": [],
      "starred-cowork-spaces": [],
      "starred-session-groups": []
    }
  },
  "coworkUserFilesPath": "/root/Claude",
  "mcpServers": {
    "kali-mcp": {
      "command": "mcp-server",
      "args": ["--server", "http://localhost:5000"]
    }
  }
}
```

bypassPermissionsGateByAccount should stay false. This is Claude Cowork's safety gate that requires confirmation before tools/files are accessed. Disabling it would let offensive tools run without per-action approval – not recommended in any environment, and especially not one wired into Metasploit, Hydra, and John.

Validate the JSON before restarting Claude Desktop – a single missing comma will silently break the whole file:

```
python3 -m json.tool ~/.config/Claude/claude_desktop_config.json
```

5. Known Issues & Fixes

5.1 CommandExecutor “expects a string, but got list”

Symptom: any tool call (nmap, gobuster, etc.) returns a 500 error, and kali-server-mcp’s console shows:

```
ValueError: CommandExecutor expects a string, but got list
```

Root cause: in /usr/share/mcp-kali-server/server.py, every tool endpoint builds its command as a list (the safe execution mode – no shell involved), but CommandExecutor.execute() contains a hard check that rejects anything that isn’t a string. This also breaks the /health endpoint, which is why all tools can show as “missing” even when installed.

Do not fix this by joining the list into a string. Several endpoints place request parameters (target, url, username, password, etc.) directly into the command. Stringifying and running via shell=True would introduce a shell-injection vulnerability. The list form already executes safely via Popen(list, shell=False) – the fix is to remove the faulty check, not to change execution mode.

Fix – in the execute() method, delete:

```
if not isinstance(self.command, str):
    raise ValueError(f"CommandExecutor expects a string, but got
{type(self.command).__name__}")

cmd_args = shlex.split(self.command)
```

Apply and restart:

```
sudo cp /usr/share/mcp-kali-server/server.py /usr/share/mcp-kali-server/server.py.bak
sudo nano /usr/share/mcp-kali-server/server.py # remove the 4 lines above, save
# Ctrl+C the running kali-server-mcp, then:
kali-server-mcp &
curl http://localhost:5000/health
```

A corrected server.py with this patch already applied was provided alongside this document.

5.2 Claude Desktop crashes: “Trace/breakpoint trap” (exit 133)

Symptom: claude-desktop exits immediately with a Trace/breakpoint trap message from launcher-common.sh.

Root cause: Electron’s chrome-sandbox helper requires SUID root permissions; this fails if permissions are wrong, the kernel restricts unprivileged user namespaces, or /usr is mounted nosuid.

6. Run the diagnostic: claude-desktop --doctor
7. Check/fix sandbox binary permissions:

```
ls -l /usr/lib/claude-desktop/node_modules/electron/dist/chrome-sandbox
sudo chown root:root /usr/lib/claude-desktop/node_modules/electron/dist/chrome-sandbox
sudo chmod 4755 /usr/lib/claude-desktop/node_modules/electron/dist/chrome-sandbox
```

8. Check the kernel setting:

```
sysctl kernel.unprivileged_userns_clone
sudo sysctl -w kernel.unprivileged_userns_clone=1
```

9. Check for a nosuid mount on /usr:

```
mount | grep " /usr "
```

Last resort only – launching with --no-sandbox works around the issue but reduces security isolation. Treat it as a diagnostic step, not a permanent fix: `claude-desktop --no-sandbox &`

5.3 “Connection refused” on localhost:5000

Symptom: tool calls fail with HTTPConnectionPool(host='localhost', port=5000) ... Connection refused.

Cause: kali-server-mcp (the Flask API) is not running.

```
ps aux | grep kali-server-mcp
ss -tlnp | grep 5000
```

If nothing is running, start it in the foreground to see errors directly:
kali-server-mcp

Once stable, confirm it's listening:
curl http://localhost:5000/

To avoid having to start it manually every session, check for and enable a systemd unit:

```
dpkg -L mcp-kali-server | grep systemd
sudo systemctl enable --now kali-server-mcp
```

5.4 Malformed claude_desktop_config.json

Symptom: python3 -m json.tool reports a parsing error (e.g. “Expecting ',' delimiter”), and Claude Desktop shows “No servers added” despite editing the config.

Cause: almost always a missing comma between key-value pairs, or a brace closed in the wrong place after manual editing in a terminal editor.

Always re-validate after editing:

```
python3 -m json.tool ~/.config/Claude/claude_desktop_config.json
```

If it prints the JSON back cleanly with no error, the file is valid.

6. Lab Network Configuration (Kali ↔ OPNsense ↔ Metasploitable2)

This section documents a routed two-subnet lab, with OPNsense as the router/firewall between Kali's network and Metasploitable2's network.

6.1 Addressing Plan

Device	Interface / Network	Address (These are Sample IP Addresses)
Kali Linux	NAT (internet/updates)	192.168.84.130/24 (DHCP)
Kali Linux	VMnet2 (lab segment)	192.168.20.10/24
OPNsense	VMnet2 interface	192.168.20.1/24
OPNsense	VMnet1 interface	192.168.10.1/24
Metasploitable2	VMnet1 (host-only)	192.168.10.20/24

6.2 Create the Second Host-Only Network

In the hypervisor's Virtual Network Editor (run with elevated privileges): add a new Host-only network, subnet 192.168.20.0/24, with "Use local DHCP service" unchecked (OPNsense will be the gateway). Name it VMnet2.

6.3 Configure Kali

10. In the Kali VM settings, set the second network adapter to VMnet2 (leave the NAT adapter as-is for internet access).

11. Identify and statically configure the new interface:

```
ip a
sudo nmcli con show
sudo nmcli con mod "Wired connection 2" ipv4.method manual ipv4.addresses 192.168.20.10/24
sudo nmcli con up "Wired connection 2"
```

12. Add a specific route to the target subnet – do not set OPNsense as the default gateway, or Kali's internet route via NAT will break:

```
sudo ip route add 192.168.10.0/24 via 192.168.20.1 dev eth1
sudo nmcli con mod "Wired connection 2" +ipv4.routes "192.168.10.0/24 192.168.20.1"
```

6.4 Configure OPNsense

13. Interfaces → Assignments: confirm both VMnet1 and VMnet2 are assigned as interfaces.

14. Assign static addresses: 192.168.10.1/24 on the VMnet1 interface, 192.168.20.1/24 on the VMnet2 interface.

15. Confirm both interfaces are configured as internal/LAN type – not WAN.

16. Firewall → Rules: add a Pass rule on each interface (source = that interface's subnet, destination = any or the opposite subnet). Newly added interfaces default to implicit deny.

6.5 Configure Metasploitable2

Point its default gateway at OPNsense so return traffic to Kali's subnet is routed correctly:

```
sudo nano /etc/network/interfaces
# set: gateway 192.168.10.1
sudo /etc/init.d/networking restart

# or, without editing the file:
sudo route add default gw 192.168.10.1
```

6.6 Verification

```
ping 192.168.10.20 # run from Kali
```

If this fails, check OPNsense → Status → System Logs → Firewall – blocked packets appear immediately and identify the missing/incorrect rule.

7. Usage Examples

With the MCP server connected, scans can be requested in natural language from the Claude Desktop chat:

- “Scan 192.168.10.20 and tell me which ports are open”
- “Run a service/version scan on 192.168.10.20” (nmap -sV)
- “Do a full port scan with OS detection on 192.168.10.20” (nmap -A -p-)
- “Enumerate directories on http://192.168.10.20 with gobuster”

Because `bypassPermissionsGateByAccount` remains false, each tool execution should prompt for confirmation before running against the target.

All rights reserved by N.M.I Raisul Bari

8. Security & Operational Notes

- Restrict all scanning/exploitation activity to the isolated lab network (Metasploitable2 on VMnet1) – do not point these tools at networks you do not own or lack explicit authorization to test.
- Leave Cowork's permission gate (bypassPermissionsGateByAccount) enabled so tool execution always requires confirmation.
- Keep kali-server-mcp bound to 127.0.0.1 unless there is a specific, access-controlled reason to expose it on the network.
- Treat claude-desktop --no-sandbox as a temporary diagnostic flag, not a permanent launch option.
- Re-apply the server.py patch (Section 5.1) after any package upgrade of mcp-kali-server, in case an update reintroduces or differs from this fix.

All rights reserved by N.M.I Raisul Bari